

MMBT3904

MMBT3904 — NPN General Purpose Transistor

Manufacturer: onsemi Package: SOT-23

General Description

The MMBT3904 is a general-purpose NPN bipolar junction transistor in a SOT-23 surface mount package. It is intended for low-power switching and amplification up to 200 mA collector current.

Features

- Collector-emitter voltage $V_{CEO} = 40\text{ V}$
- Continuous collector current $I_C = 200\text{ mA}$
- High current gain: $h_{FE} 100$ to 300 at $I_C = 10\text{ mA}$
- SOT-23 (SC-59) package, $2.9 \times 1.6\text{ mm}$ body

Pin Configuration

Pin assignment for the SOT-23 package (top view).

Pin	Name	Function
1	Base	Control input
2	Emitter	Emitter, usually grounded
3	Collector	Switched output

Absolute Maximum Ratings

Stresses beyond these values may cause permanent damage to the device.

Symbol	Parameter	Value	Unit
V_{CEO}	Collector-emitter voltage	40	V
V_{CBO}	Collector-base voltage	60	V
V_{EBO}	Emitter-base voltage	6.0	V
I_C	Collector current, continuous	200	mA
P_{tot}	Total power dissipation	350	mW
T_j	Junction temperature	150	degC

Electrical Characteristics

Typical values at TA = 25 degC unless otherwise noted.

Symbol	Parameter	Test condition	Min	Typ	Max	Unit
VCE(sat)	Collector-emitter saturation	IC = 50 mA, IB = 5 mA	-	0.15	0.30	V
hFE	DC current gain	VCE = 1 V, IC = 10 mA	100	-	300	-
fT	Transition frequency	VCE = 20 V, IC = 10 mA	250	300	-	MHz
Cobo	Output capacitance	VCB = 5 V, f = 1 MHz	-	4.0	-	pF

Thermal Characteristics

Symbol	Parameter	Value	Unit
RthJA	Thermal resistance junction-ambient	357	K/W
RthJL	Thermal resistance junction-lead	83	K/W

Package Dimensions

Outline drawing for the SOT-23 package. All dimensions in millimetres.

Dimension	Min	Nom	Max	Unit
A (body length)	2.80	2.90	3.04	mm
B (body width)	1.20	1.30	1.45	mm
C (height)	0.89	1.00	1.12	mm
e (lead pitch)	-	1.90	-	mm

Ordering Information

Order code	Package	Marking	Packing
MMBT3904	SOT-23	1A	Reel, 3000 pcs
MMBT3904LT1G	SOT-23	1A	Reel, 3000 pcs

Document revision 1.0. Values are typical at TA = 25 degC. Verify against the latest datasheet from onsemi before design release.